

Curriculum Vitae

Employment

2025–present **Postdoc**, *Max Planck Institute for Mathematics in the Sciences (MPI-MiS), Leipzig.*

Education

2022–2025 **Ph.D. in Mathematics**, *IMB, Université de Bordeaux.*

2020–2022 **Master in Mathematics**, *Indiana University Bloomington.*

2017–2020 **Master in Mathematics**, *Nankai University.*

2013–2017 **Bachelor in Mathematics and Applied Mathematics**, *Beijing Institute of Technology.*

Research Interests

My research centers on surfaces (real two-dimensional manifolds), with an emphasis on the connections between geometry, combinatorics, and dynamics. In particular, I study the geometry and dynamics of translation surfaces and flat cone surfaces, as well as structures on their moduli spaces.

Publications and Preprints

Siegel–Veech Measures of Convex Flat Cone Spheres, Kai Fu, *Compositio Mathematica*, 162(7) (2026), 1507–1573. [arXiv](#), [DOI](#) .

Uniform Length Estimates for Trajectories on Flat Cone Surfaces, Kai Fu, To appear in *Annales de l'Institut Fourier*. [arXiv](#) .

Bounds on saddle connections for flat spheres, Kai Fu, Guillaume Tahar, *Journal of Topology*, 19(2) (2026). [arXiv](#), [DOI](#) .

In Preparation

L^∞ -Delaunay cells and linear subvarieties in the moduli space of Abelian differentials, with Vincent Delecroix and Bradley Zykoski.

Point Processes and Gap Statistics for Translation Surfaces, with Anthony Sanchez.

Programming Project

I use computational methods in my research, often with SageMath. Currently, I am working with Vincent Delecroix on the computation of real linear subvarieties in the moduli spaces of Abelian differentials. More precisely, we develop algorithms to verify that a given germ defines a real linear subvariety. We also compute the boundary of a real linear subvariety in the multi-scale compactification. This project is being implemented in the package veerer.

Talks

- Jul 15, 2026 **Geometry and Groups Seminar**, *Heidelberg University*, Heidelberg, Germany.
- Jul 09, 2026 **Oberseminar Differentialgeometrie**, *Max Planck Institute for Mathematics*, Bonn, Germany.
- Jun 02, 2026 **CMI-HIMR Summer School on Random Geometries and Random Matrices**, *University of Bristol*, (lightning talk), Bristol, United Kingdom.
- Nov 25, 2025 **The floor to the junior researchers of Plato's tetrahedron's research network**, *Université de Montpellier*, Montpellier, France.
- Nov 18, 2025 **UCL/KCL Junior Geometry Seminar**, *King's College*, London, United Kingdom.
- Oct 08, 2025 **Géométrie et dynamique dans les espaces de modules**, *Institut Henri Poincaré (IHP)*, Paris, France.
- Sep 25, 2025 **Fifth GALS meeting**, *Institut Henri Poincaré (IHP)*, Paris, France.
- Aug 25, 2025 **Seminar**, *School of Mathematics, Sun Yat-sen University*, Guangzhou, China.
- Aug 15, 2025 **Dynamical Systems Seminar**, *Shanghai Institute for Mathematics and Interdisciplinary Sciences*, Shanghai, China.
- Jul 29, 2025 **Seminar**, *Chern Institute of Mathematics*, Tianjin, China.
- Jul 17, 2025 **The 5th Korea-France Conference in Mathematics**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.
- Jun 27, 2025 **Séminaire de Géométrie**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.
- Nov 08, 2024 **Séminaire de Systèmes Dynamiques**, *Institut de mathématiques de Toulouse*, Toulouse, France.
- Nov 07, 2024 **Groupe de travail**, *Institut de mathématiques de Toulouse*, Toulouse, France.
- Oct 04, 2024 **Geometry Topology Grad and Postdoc Seminar**, *University of Toronto*, online.
- Sep 10, 2024 **Riemann Surfaces: Random, Flat, and Hyperbolic Geometry**, *MFO*, (Lightning talk), Oberwolfach, Germany.
- Oct 17, 2023 **Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUX**, *Le Laboratoire de Mathématiques et Applications, Université de Poitiers Poitiers*, (Short talk), Poitiers, France.
- Jun 08, 2023 **Student Seminar**, *YMSC, Tsinghua University*, Beijing, China.
- Mar 16, 2023 **Reading group on mapping class groups**, *Institut de mathématiques de Bordeaux*, Bordeaux, France.

Visiting

- Jul, 2026, MPIM Bonn (1 week)
- Aug, 2025, School of Mathematics, Sun Yat-sen University (1 week)
- Jul, 2025, Chern Institute of Mathematics (1 week)
- Jun, 2023, YMSC, Tsinghua University (3 weeks)

Organization

- Mar, 2024 Co-organization of the School on flat surfaces and interactions, Le Teich
- Jan - May, 2023 Co-organization of the Reading group on mapping class groups, Bordeaux

Conference Participation

- Jun, 2026 Geometry, Dynamics, and Computer-Assisted Proofs, a conference in honor of Rich Schwartz, Heidelberg University, Heidelberg
- Jun, 2026 CMI-HIMR Summer School on Random Geometries and Random Matrices, University of Bristol, Bristol
- Sep, 2025 Fifth GALS meeting, IHP, Paris
- Jul, 2025 The 5th Korea-France Conference in Mathematics, IMB, Bordeaux
- Nov, 2024 Riemann Surfaces: Random, Flat, and Hyperbolic Geometry, MFO, Oberwolfach
- Jun, 2024 New Frontiers in Parabolic Dynamics and Renormalization, Department of Mathematics, University of Bologna, Bologna
- Jun, 2024 Rencontre GALS, Institut Camille Jordan, Lyon
- Mar, 2024 School on flat surfaces and interactions, Le Teich
- Jan, 2024 Sage Days 125: combinatorics, number theory, dynamical systems and geometry, Le Teich
- Jan, 2024 Research School - Hodge Theory and o-minimality, CIRM, Marseille
- Dec, 2023 Renormalization and Visualization in Geometry, Dynamics and Number Theory, CIRM, Marseille
- Nov, 2023 The floor to the junior researcher of Plato's network Institut Fourier, Grenoble
- Oct, 2023 Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUX, Université de Poitiers, Poitiers
- Oct, 2023 Probability and Geometry in, on and of non-Euclidian spaces, CIRM, Marseille
- Jul, 2023 Renormalization and Visualization for packing, billiard and surfaces, CIRM, Marseille
- Feb, 2023 Sage Days 117: combinatorics, number theory, dynamical systems and geometry, Le Teich
- Sep, 2022 Combinatorics, Dynamics and Geometry on Moduli Spaces, CIRM, Marseille
- May, 2022 Structures on Surfaces, CIRM, Marseille
- Apr, 2022 Mini-rencontre ANR MoDiff, LaBRI, Université de Bordeaux, Bordeaux
- Jun, 2021 CMI-HIMR Dynamics and Geometry Online Summer School, University of Bristol

Languages

Chinese native

English fluent

French basic